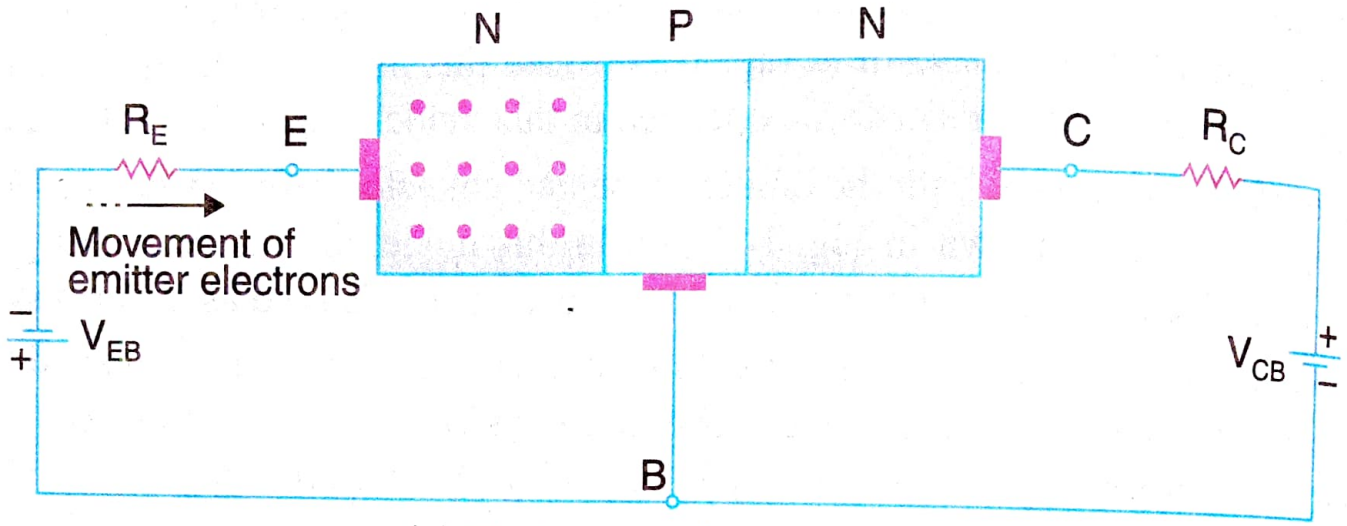
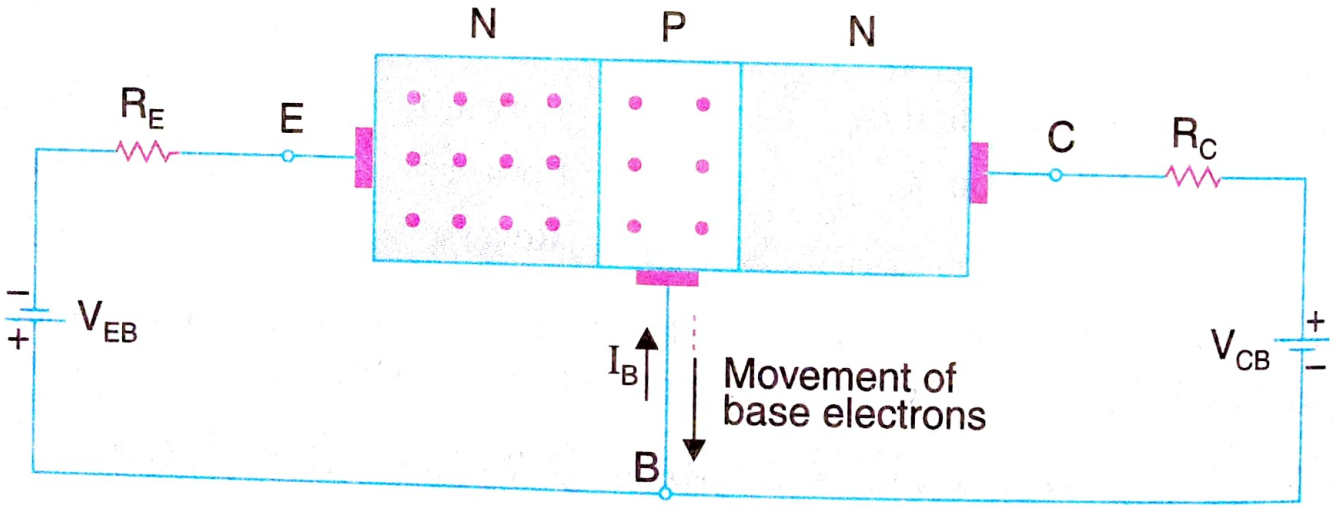
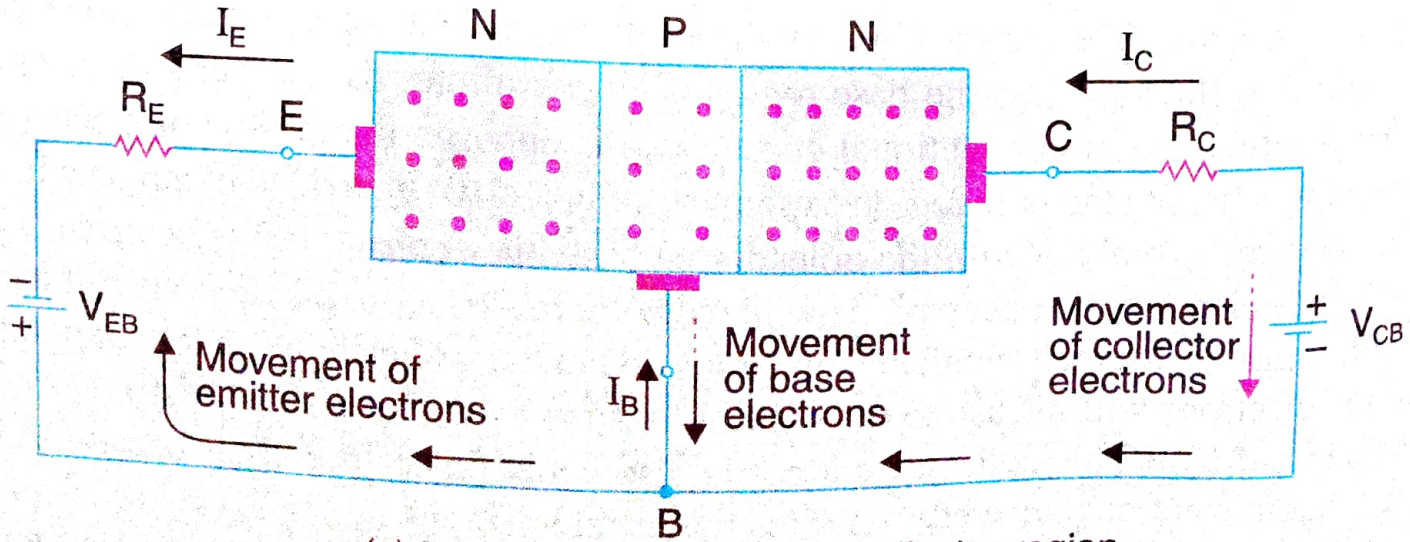


(*a*) Electrons in the emitter region.



(*b*) Electrons in the base region.



(*c*) Electrons entered in the collector region.

Fig. 14.10.

14.8. Operation of an NPN Bipolar Junction Transistor

Fig. 14.10 (*a*) shows an NPN transistor biased in forward-active mode. *.i.e.*, the emitter-base of a transistor is forward-biased and collector-base junction is reverse-biased. The emitter-base junction is forward biased only if V_{EB} is greater than barrier potential which is 0.7 V for silicon and 0.3 V for germanium transistors. The forward bias on the emitter-base junction causes the free electrons in the N-type emitter to flow towards the base region. This constitutes the emitter current (I_E). It may be noted that the direction of conventional current (I_C) is opposite to the flow of electrons. Therefore electrons, after reaching the base region as shown in Fig. 14.10 (*b*), tend to combine with the holes. If these free electrons combine with the holes in the base, they constitute base current (I_B). However, most of the free electrons do not combine with the holes in the base. This is because of the fact that the base width is made extremely small and electrons do not get sufficient holes for recombination. Thus most of the electrons will diffuse to the collector region as shown in Fig. 14.10 (*c*) and constitutes collector current (I_C). This collector current is also called injected current because this current is produced due to electrons injected from the emitter region.

It will be interesting to know that there is another component of collector current due to the thermally generated carriers. This current component is called reverse saturation current and is quite small.

Notes 1

14.9. Operation of a PNP Bipolar Junction Transistor

The operation of a PNP transistor is similar to that of an NPN transistor. However, the current within a PNP transistor is due to the movement of holes, whereas in an NPN transistor it is due to the movement of free electrons. Fig. 14.11 shows a PNP transistor with its emitter-base junction as forward-biased and collector-base junction as reverse biased. The forward bias on the emitter-base junction causes the holes in the emitter region to flow towards the base region. This constitutes the emitter current (I_E). The holes, after reaching in the base region, tend to combine with the electrons. If these holes combine with the electrons in the base, they constitute base current (I_B). However, most of the holes do not combine with the electrons in the base region. It is due to the fact that base width is made extremely small and holes do not get sufficient electrons for recombination. Thus most of the

holes diffuse to the collector regions and constitute collector current (I_C). This collector current is called injected current because this current is produced due to the holes injected from the emitter region.

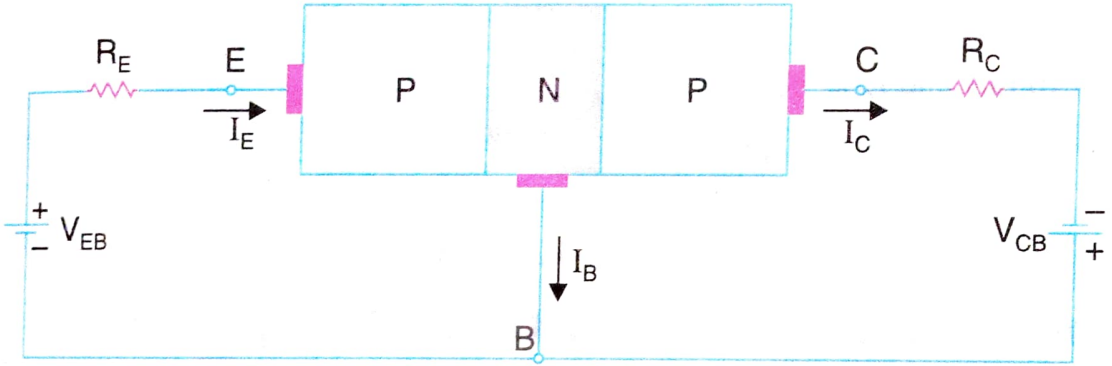


Fig. 14.11. Operation of a PNP transistor.

There is another small component of collector current due to thermally generated carriers. This current component is called reverse saturation current. The conventional directions of emitter, base and collector currents are shown by I_E, I_B and I_C respectively in Fig. 14.11.

11.2. Formation of Depletion Layer in a PN Junction

PN junction diode working

Figure 11.2 (*a*) shows a semiconductor consisting of a PN junction, which has been just formed. It may be noted that this entire sample is a single crystal. Its left half is a P-type and the right half is N-type. The P-region has holes (as majority carriers) and negatively charged impurity atoms, called negative ions (or acceptor ions). The N-region has free electrons (as majority carriers) and positively charged impurity atoms, called positive ions (or donor ions). The minority carriers in the P-and N-regions are not shown just for simplicity.

We know that holes and electrons are mobile charges, and therefore, are known as moble charge carriers. On the other hand, positive and negative ions are immobile charges and therefore, do not take part in the conduction. The sample, as a whole, is electrically neutral and so are the P- and *N*-regions, considered separately. Thus in the *P*-region, the total positive charge on the holes is equal to the total negative charges on free electrons and immobile ions. Similarly, in the N-region, the total negative charge on free electrons is equal to the total positive charge on holes and immobile ions.

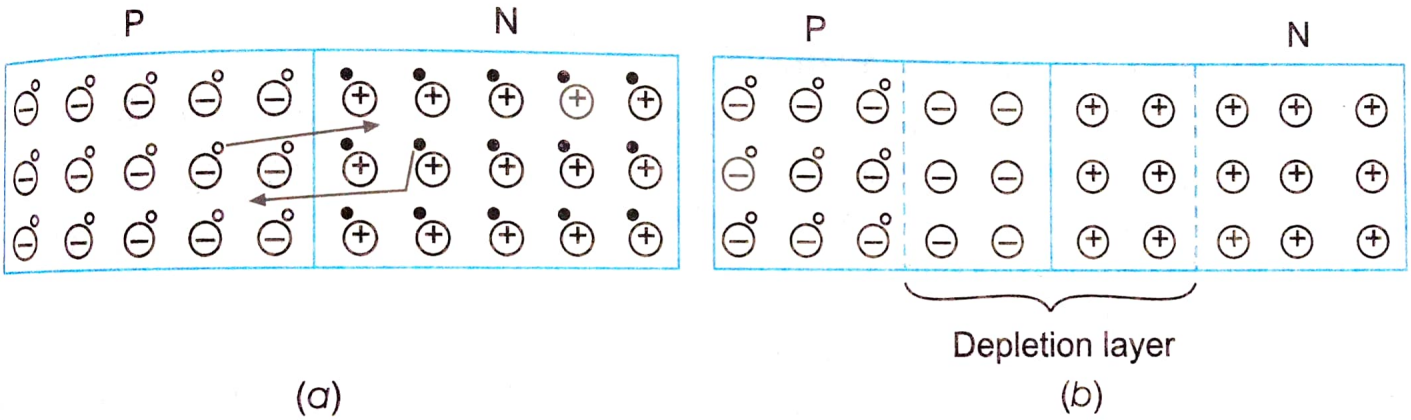


Fig. 11.2. Formation of deplection layer in a PN junction.

It may be noted that no external voltage has been connected to the PN junction of Fig. 11.2. As soon as the junction is formed, the conduction and valence bands of P- and N-type materials overlap. As a result of this the following processes take place:

1. The holes, from the P-region diffuse to the N-region, where they combine with the free electrons.
2. The free electrons, from the N-region diffuse to the P-region, where they combine with holes.
3. The diffusion of holes (from P-region to N-region) and free electrons (from N-region to P-region) takes place due to the reason that there is a difference of concentrations in the two regions *i.e.*, the P-region has more number of holes, whereas N-region has more number of free electrons. This difference in concentration creates a concentration gradient across the junction. It results in the diffusion of mobile charge carries across the junction. Moreover, the electrons and holes move at random in all directions due to thermal energy. In doing so, some carriers manage to crossover the junction.

4. The diffusion of holes and free electrons across the junction takes place for a short time. After a few recombination of holes and free elections, in the vicinity of the junction, a restraining force is automatically set up. This force is produced due to depletion region, which exists on either side of the junction. As a result of this, further diffusion of holes and free electrons from one region to the other is stopped by this depletion layer. The formation of this depletion is explained below:

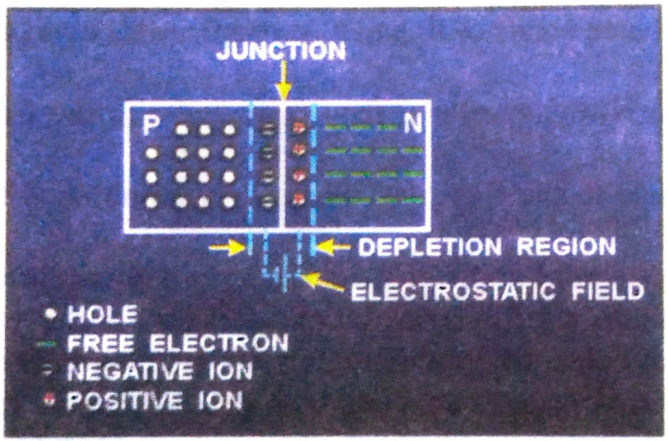
We know that as soon as the PN junction is formed, some of the holes in P-region and the free electrons in the N-region diffuse in each other and disappear due to recombination. In this process, the negative acceptor ions in the P-region and positive donor ions in the N-region are left uncovered (or uncompensated) in the vicinity of junction as shown in Fig. 11.2 (*b*). The additional holes, trying to diffuse to the N-region, are repelled by the uncovered positive charge of the donor ions. Similarly, the electrons, trying to diffuse into the P-region, are repelled by the uncovered negative charge of the acceptor ions. As a result of this, the further diffusion of free electrons and holes across the junction is stopped.

The region containing the uncovered accepter and donor ions, in the vicinity of the junction, is called depletion region. Since this region has immobile (or fixed) ions, which are electrically charged, therefore, the depletion region is also known as space-charge region. Moreover, as the uncovered charges within the depletion region exists in the form of parallel rows or plates of opposite charges, therefore, it is known as depletion layer. The depletion layer behaves like an insulator. Because of the presence of rows of fixed charges, the depletion layer possesses capacitance. It will be discussed in detail in Art. 11.12.

It will be interesting to know that the width of depletion layer depends upon the doping level of the impurity in *N*-type or *P*-type semiconductor. The higher the doping level, the thinner will be the depletion layer and vice-versa. It is due to the fact that a highly doped PN junction contains a large number of electrons and holes. Because of this, a diffusing charge carrier (either free electron or hole) has not to travel across the junction for recombination.

11.3. Junction or Barrier Voltage

We have already discussed in the last article that the depletion layer of a PN junction has no mobile charge curriers. But it contains fixed rows of oppositely charged ions on its two sides. Because of this charge separation, an electric potential (designated as V_B) is established across the junction, even when the junction is not connected to any external voltage source as shown in Fig. 11.3. This electric potential is called *junction* or *potential barrier.*



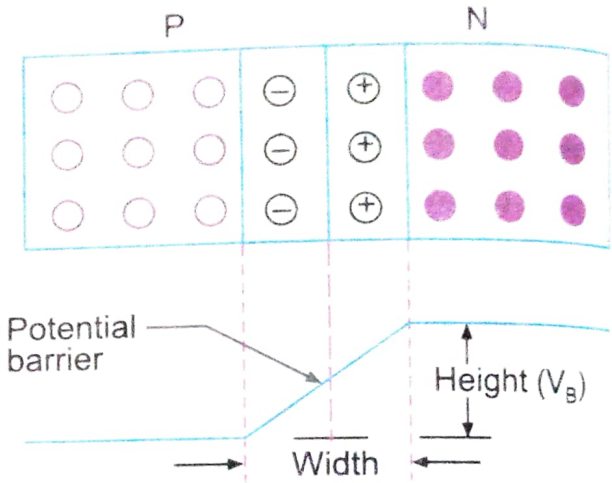


Fig. 11.3. Junction or barrier voltage.

It will be interesting to know that this potential barrier exerts a repelling force on the mobile charge carriers, trying to crossover the junction. This force stops the mobile charge carriers to crossover the junction, unless the energy is supplied from an external source. At room temperature, (*i.e.*, at 300 K), the value of V_B

$V_B = 0.6$ V for silicon and 0.2 V for germanium.